

Aditya Sowdagor

Machine Learning Researcher · Data Science & AI Specialist

📍 Dhaka, Bangladesh 📞 01956215604 ✉ sowdagor.aditya@gmail.com

🌐 [linkedin.com/in/aditya-sowdagor](https://www.linkedin.com/in/aditya-sowdagor) 🏠 github.com/CollabwithAdi 📞 0009-0000-5274-9118

ACADEMIC OBJECTIVE & SUMMARY

Highly motivated Computer Science graduate with a specialized focus in Data Science, Machine Learning, and AI. Proven research capabilities demonstrated through the development of multimodal deep learning models for speech restoration, autonomous robotics, and AI-powered IoT wearables. Well-versed in Institutional Review Board (IRB) protocols, ethical clearance processes, and the secure handling of sensitive medical data. Experienced in end-to-end model training, data pipeline architecture, and cloud infrastructure (Microsoft Azure & RHCSA Certified). Recipient of World Bank pre-seed funding for AI R&D initiatives. Actively seeking Graduate Research Assistant opportunities to contribute to cutting-edge academic research in Applied Artificial Intelligence, NLP, Multimodal Systems, and Healthcare Tech.

TECHNICAL SKILLS

- **Machine Learning & AI:** Model Training & Evaluation, Audio Signal Processing, Computer Vision, NLP, Deep Learning
- **Programming Languages:** Python, C, Java, SQL
- **Cloud & DevOps:** Microsoft Azure AI, Microsoft Azure Administrator, Linux Administration (RHCSA), Git/GitHub
- **Academic & Research Tools:** LaTeX, Overleaf, Zotero, Mendeley, Jupyter Notebook, Google Colab
- **Research & Compliance:** IRB Protocols, Ethical Clearance Processes, Medical Data Privacy, Experimental Design
- **Architecture & Hardware:** Blockchain Integration, Microcontrollers, IoT Sensor Integration, Wearables

RESEARCH & PROFESSIONAL EXPERIENCE

Founder & Lead AI Architect

Shohochor

September 2025 – Present

Dhaka, Bangladesh

- Lead the research and development of an AI-powered smartwatch designed to protect children by detecting environmental bullying and emotional stress.
- Train and deploy machine learning models for real-time ambient audio processing, utilizing culturally-aware speech analysis to trigger instant guardian alerts.
- Architect an end-to-end IoT system integrating GPS location tracking, health monitoring sensors, a parent-facing mobile app, and a school management dashboard.
- Secure and allocate R&D funding, bridging the gap between cutting-edge wearable technology and child safety.

Startup Graduate

Accelerating Bangladesh

July 2025 – September 2025

Dhaka, Bangladesh

- Graduated from a highly competitive 3-month hybrid startup acceleration program focused on transitioning AI research into viable tech products.
- Received advanced training on product-market fit, scalable infrastructure deployment, and technical business development for Shohochor.

Mentee (Pre-Seed Fund Winner)

University Innovation Hub Program - UIHP

November 2024 – April 2025

Dhaka, Bangladesh

- Secured prestigious research and development funding from the World Bank to support the scaling of AI-driven technology initiatives (Shohochor).
- Collaborated with industry experts and technical advisors to refine AI deployment strategies and system architectures.

- Developed **WORKIEX**, a scalable full-stack platform designed to optimize job search and transaction matching for freelancers.
- Designed structured database architectures and implemented user-friendly workflows under the guidance of Microsoft engineering mentors.
- Recognized among the **Top 30 out of around 200 interns nationwide** based on the successful deployment and high-quality technical contributions.

EDUCATION

B.Sc. in Computer Science and Engineering (Data Science)
United International University

June 2021 – 2026

- **Award:** Champion - Final Year Design Project

International Advanced Level (IAL)
British Council Bangladesh (Pearson Edexcel)

January 2018 – November 2020

International GCSE (IGCSE)
British Council Bangladesh (Pearson Edexcel)

January 2017 – January 2018

PUBLICATIONS & MANUSCRIPTS

- [1] **Sowdagar, A.**, [Co-Authors]. *[Title Withheld]*. [Target Journal/Conference]. **[In Submission Process]**
- [2] **Sowdagar, A.**, [Co-Authors]. *[Title Withheld]*. [Target Journal/Conference]. **[In Submission Process]**
- [3] **Sowdagar, A.**, [Co-Authors]. *[Title Withheld]*. [Target Journal/Conference]. **[In Submission Process]**
- [4] **Sowdagar, A.**, [Co-Authors]. *[Title Withheld]*. [Target Journal/Conference]. **[In Submission Process]**

TECHNICAL PROJECTS

DigiDignity – AI-Integrated Blockchain Identity System (Live Deployment)
Architecture

Blockchain, AI, System

- Designed an AI-integrated blockchain framework (AIBI) to solve the global identity crisis for displaced populations.
- Engineered a decentralized architecture to provide secure, portable, and verifiable digital identities for refugees, recognized as a National Finalist at the 2025 Blockchain Olympiad Bangladesh.

IntelliNurse – Autonomous Patient Assistance Rover

Microcontrollers, IoT, Sensor Integration

- Designed and prototyped an autonomous robotics system for elderly patient monitoring, awarded 3rd Runner-Up in the Microprocessor and Microcontroller Laboratory.
- Implemented a tri-modal navigation system featuring Voice, Gesture, and Manual controls to ensure high accessibility for users.
- Integrated real-time health diagnostics (Blood Pressure, SpO2) and sensor-based activity tracking.

IntelliSense – AI-Powered Smart Wheelchair System (Video Demo)

Hardware & IoT, C

- Engineered a smart wheelchair integrating gesture recognition and natural language voice controls, awarded 1st Runner-Up in Electronics Laboratory.
- Developed and integrated advanced sensor arrays for real-time anomaly detection, including fire alerts, body fall detection, and obstacle avoidance algorithms.

CitySync – Civic Management Platform (Video Demo)

Object-Oriented Programming (Java)

- Created an object-oriented software platform aimed at diminishing the communication gap between government representatives and local citizens, awarded 1st Runner-Up in Advanced OOP.

HONORS, CERTIFICATIONS & FEATURES

- **Certifications:** Red Hat Certified System Administrator (RHCSA) — Microsoft Azure AI — Microsoft Azure Administrator — Aspire Leaders Program
- **Blockchain Olympiad Bangladesh (BCOLBD):** National Finalist 2025 (Project: DigiDignity)
- **United International University Awards:** Champion (Final Year Design Project), 1st Runner-Up (Advanced OOP, Electronics Lab, System Analysis Lab), 3rd Runner-Up (Microprocessor and Microcontroller Lab)
- **Media Feature:** Featured Tech Startup (*Shohochor*) on [Accelerating Bangladesh](#)